

Population: In any statistical investigation the group of individual or objects under study is known as population and size of population is denoted by 'N'.

Sample: A finite subset of population is called a sample and size of sample is denoted by 'n'.

The process of selecting a sample from the population is called **sampling**.

If the sample size n is greater than 30 then it is called **large sample** otherwise it is called **small sample**.

Parameter & Statistic: Any statistical measure based on entire population is called **parameter**.

Ex: Population mean (μ)

Population standard deviation (σ)

Population proportion (P)

↳ % value of an attribute

②

Any statistical measure computed from a sample is called **statistic**.

Ex: Sample mean ($\bar{x}$)

Sample standard deviation (s)

Sample proportion (p)

Sampling Distribution:

Let Population size = N & Sample size = n .

The number of samples of size $n = N C_n = k$ (say)

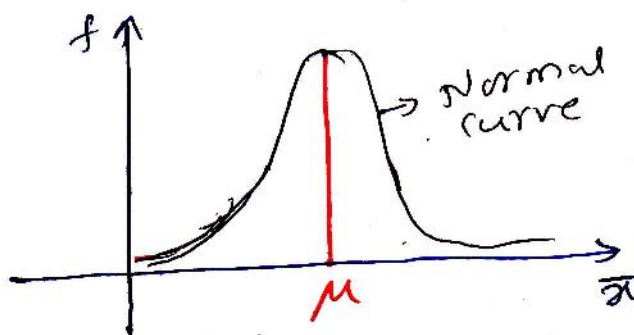
For each of these samples, compute a statistic (Mean, s.d, etc.) which may vary from sample to sample. The collection of all the value of statistic along with their frequency constitute sampling distribution.

Ex Population mean = μ

Sampling mean for different samples = $\bar{x}$

$= \{ \bar{x}_1, \bar{x}_2, \bar{x}_3, \dots, \bar{x}_n \}$

The sampling distribution of means is given by



③

In the sampling distribution discussed above, if Statistic is mean then the distribution is called sampling distribution of means. Then,

Population mean
↑

$$\text{mean of distribution} = \text{Mean}(\bar{x}) = \mu$$

Standard deviation of distribution also called standard error (S.E.) = $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$

Similarly, in the sampling distribution discussed above, if Statistic is Proportion
[Proportion $\rightarrow$ Ratio of no. of objects possessing a characteristic to total no. of objects in a population (sample)]

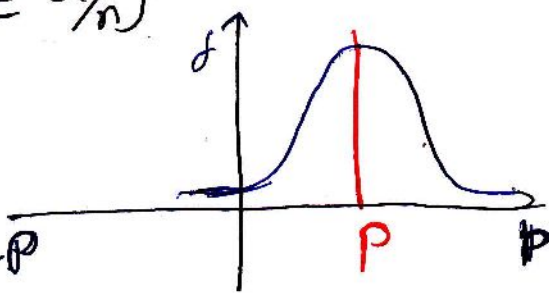
then the distribution is called sampling distribution of Proportions.

If Population Proportion = $P (= \frac{X}{N})$ &

Sample proportion = $p (= \frac{x}{n})$

$$\text{Mean}(p) = M_p = P$$

$$\text{S.E.}(p) = \sigma_p = \sqrt{\frac{pq}{n}}, \quad q = 1 - p$$



Sampling distribution of difference of means

Let $\bar{x}_1$ & $\bar{x}_2$ denote means of two samples of sizes n_1 & n_2 drawn from two populations having means M_1 & M_2 and standard deviations σ_1 & σ_2 , then

$$\text{Mean}(\bar{x}_1 - \bar{x}_2) = M_{\bar{x}_1 - \bar{x}_2} = M_1 - M_2$$

$$\text{S.E.}(\bar{x}_1 - \bar{x}_2) = \sigma_{\bar{x}_1 - \bar{x}_2} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

Similarly for difference of proportions,

$$\text{Mean}(p_1 - p_2) = M_{p_1 - p_2} = P_1 - P_2$$

$$\text{S.E.}(p_1 - p_2) = \sigma_{p_1 - p_2} = \sqrt{\frac{P_1 Q_1}{n_1} + \frac{P_2 Q_2}{n_2}}, \quad Q_1 = 1 - P_1, \quad Q_2 = 1 - P_2$$

Testing of Hypothesis

In order to arrive at a decision regarding the population through a sample of the population, we have to make certain assumption referred to as **hypothesis** which may or may not be true.

Null hypothesis: The hypothesis formulated for the purpose of its rejection under the assumption that it is true & is denoted by H_0 .

An hypothesis which is complementary to the null hypothesis is called **Alternative hypothesis** and is denoted by H_1 .

Errors in Testing of hypothesis

The decision to accept or reject the null hypothesis H_0 is made on the basis of the information supplied by the sample data and there is always a chance of making error.

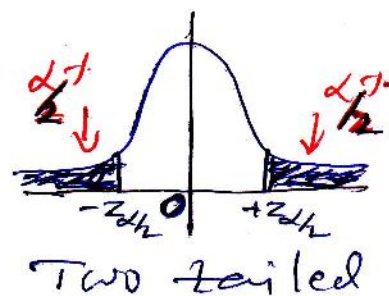
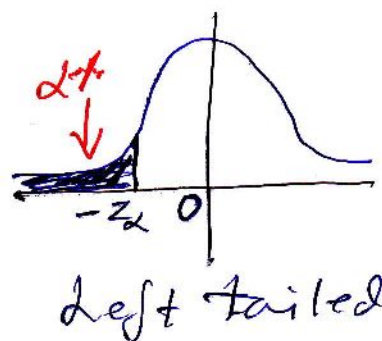
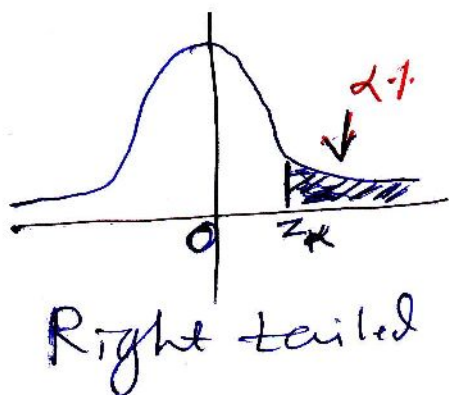
The error of rejecting H_0 when H_0 is true is known as Type I error.

The error of accepting H_0 when H_0 is not true is known as Type II error.

Actual	Decision from Sample	
	Accept H_0	Reject H_0
H_0 is true	correct decision	wrong decision (Type -I error)
H_0 is not true	wrong decision (Type II error)	correct decision

Level of Significance & Critical Region

The maximum probability of committing type I/II error is called level of significance & is denoted by α , and the region corresponding to rejection of null hypothesis is called critical region.



For $\alpha = 5\% = 0.05$

Right tailed $\rightarrow P[Z \leq z_\alpha] = 0.95 \Rightarrow 0.5 + A(z_\alpha) = 0.95$
 $\Rightarrow A(z_\alpha) = 0.45 \Rightarrow \boxed{z_\alpha \approx 1.645}$

Left tailed $\rightarrow P[Z \geq -z_\alpha] = 0.95 \Rightarrow \boxed{z_\alpha = -1.645}$

Two tailed $\rightarrow P[-z_{\alpha/2} \leq Z \leq z_{\alpha/2}] = 0.95 \Rightarrow 2P[0 \leq Z \leq z_{\alpha/2}] = 0.95$
 $\Rightarrow A(z_{\alpha/2}) = 0.475 \Rightarrow \boxed{z_{\alpha/2} \approx 1.96}$

Nature of Test	Level of Significance (α)		
	$\alpha = 0.01 (1\%)$	$\alpha = 0.05 (5\%)$	$\alpha = 0.1 (10\%)$
Two	2.58	1.96	1.645
Right	2.33	1.645	1.28
Left	-2.33	-1.645	-1.28

Procedure for Testing of Hypothesis

- Step ① Set up Null hypothesis H_0
- Step ② Set up Alternative hypothesis H_1 & decide nature of test
- Step ③ Choose level of significance (α)
- Step ④ Compute test statistic
- $$Z = \frac{\bar{x} - E(\bar{x})}{S.E(\bar{x})}$$
- Step ⑤ Compare calculated value of Z with Z_α (or $Z_{\alpha/2}$)
- Step ⑥ If Z_{cal} is outside critical region then H_0 may be accepted or else H_0 ~~may be~~ ^{is} rejected.
- Step ⑦ Write down conclusion about test

NOTE:

- ① Test for Mean: $Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$
- ② Test for equality of means: $Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$
- ③ Test for Proportion: $Z = \frac{p - P}{\sqrt{PQ/n}}$
- ④ Test for equality of Proportions: $Z = \frac{p_1 - p_2}{\sqrt{\frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}}}$

Examples

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Ex (1) The mean life of a sample of 100 bulbs produced by a company is computed to be 1570 hrs with SD of 120 hrs. If μ is the life of all bulbs produced by the company, test the hypothesis $\mu = 1600$ hrs, using level of significance 0.01

Solⁿ

H_0 : Population mean $\mu = 1600$

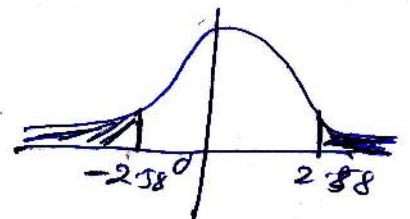
H_1 : Population mean $\mu \neq 1600$ ($\mu < 1600$ or $\mu > 1600$)
(Two-tailed)

$$\text{Test statistic: } z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{1570 - 1600}{120/\sqrt{100}} = -2.5$$

$\therefore z_{\text{cal}} = -2.5 \notin \text{critical region}$

$\therefore H_0$ may be accepted

i.e. Mean life of bulbs produced by the company is 1600 hrs.



Ex (2) A coin was tossed 400 times & head turned up 216 times. Test the hypothesis at 5% level of significance that coin is not biased.

Solⁿ P: Population proportion of heads = $\frac{1}{2}$
 p = Sample proportion of heads = $\frac{216}{400} = 0.54$

def

H_0 : coin is unbiased: $P = 0.5$

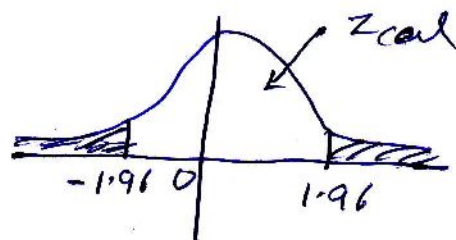
H_1 : coin is not unbiased: $P \neq 0.5$ (Two tailed)

Test statistic: $Z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} = \frac{0.54 - 0.5}{\sqrt{\frac{0.5 \times 0.5}{400}}} = 1.6$

$\therefore Z_{cal} \notin$ critical region

$\therefore H_0$ is accepted

$\therefore$ coin is unbiased.



Ex (2) A die was thrown 1200 times & number 6 was obtained 236 times. Can the die be considered as fair at 0.01 level of significance

Solⁿ P: population proportion of getting 6 = $\frac{1}{6}$
 p : sample proportion of getting 6 = $\frac{236}{1200} = 0.1967$

def H_0 : Die is fair (unbiased): $P = \frac{1}{6}$

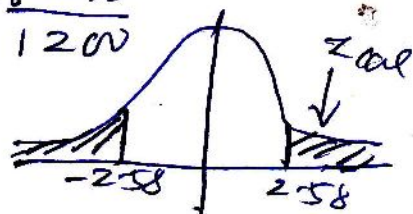
H_1 : Die is not fair (biased): $P \neq \frac{1}{6}$ (Two tailed)

Test statistic: $Z = \frac{p - P_0}{\sqrt{\frac{P_0 Q_0}{n}}} = \frac{0.1967 - \frac{1}{6}}{\sqrt{\frac{\frac{1}{6} \times \frac{5}{6}}{1200}}} = 2.78$

$\therefore Z_{cal} \in$ critical region

$\therefore H_0$ is rejected

$\therefore$ The die is not fair.



Ex (4) A die was thrown 9000 and a throw of 5 or 6 was obtained 3240 times. Do the data indicates that die is unbiased (10)

Solⁿ P: population proportion of getting 5 or 6
 $= \frac{1}{6} + \frac{1}{6} = \frac{1}{3}$

p: sample proportion of getting 5 or 6
 $= \frac{3240}{9000} = 0.36$

Let H_0 : Die is unbiased: $P = \frac{1}{3}$

H_1 : Die is not unbiased $P \neq \frac{1}{3}$ (Two tailed)

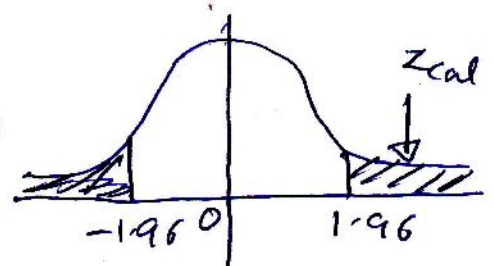
$$\text{Test statistic: } Z = \frac{p - P}{\sqrt{PQ/n}} = \frac{0.36 - \frac{1}{3}}{\sqrt{\frac{\frac{1}{3} \times \frac{2}{3}}{9000}}} = 5.37$$

Since level of significance not given,
assume $\alpha = 5\% = 0.05$

Since $Z_{cal} \in$ critical region

$\therefore H_0$ may be rejected

$\therefore$ Die is biased.

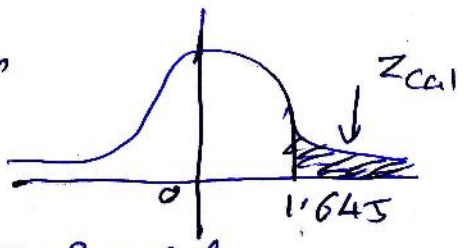


Ex 5 A random sample of 100 recorded deaths on past year showed an average life span of 71.8 years. Assuming a population S.D. of 8.9 years, does the data indicate that average span today is greater than 70 years. Use 0.05 level of significance

Soln H_0 : Average life span is 70: $M = 70$
 H_1 : Average life span is greater than 70
: $M > 70$ (Right tailed)

Test statistic: $Z = \frac{\bar{x} - M}{\sigma/\sqrt{n}} = \frac{71.8 - 70}{8.9/\sqrt{100}} = 2.022$

Since $Z_{cal} \in$ critical region
 $\therefore H_0$ may be rejected



$\therefore$ Avg. life span is greater than 70 years.

Ex 6 A sample of 100 bulbs produced by a company A showed a mean life of 1190 hrs & S.D. of 90 hrs. Also a sample of 75 bulbs, produced by a company B showed a mean life of 1230 hrs & a S.D. of 120 hrs. Is there a difference between the mean life of the bulbs produced by two companies at (i) 5% level of significance (ii) 1% level of significance

Solⁿ Given $n_1 = 100$, $\bar{x}_1 = 1190$, $\sigma_1 = 90$
 $n_2 = 75$, $\bar{x}_2 = 1230$, $\sigma_2 = 120$

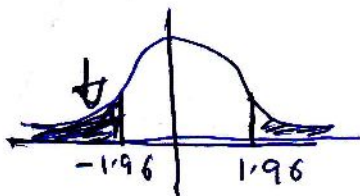
Let

H_0 : There is no significant difference between the mean life of bulbs produced by two companies: $\mu_1 = \mu_2$

H_1 : There is a significant difference between the mean life of bulbs produced by two companies: $\mu_1 \neq \mu_2$ (Two-tailed)

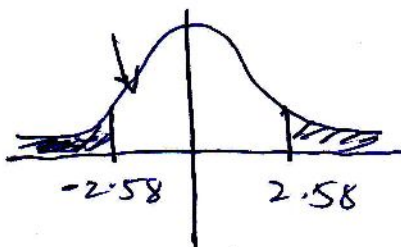
Test statistic:
$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{1190 - 1230}{\sqrt{\frac{90^2}{100} + \frac{120^2}{75}}} = -2.42$$

① At 5% level of significance



$Z_{cal} \in \text{critical region}$
 $\therefore H_0$ may be rejected

② At 1% level of significance



$Z_{cal} \notin \text{critical region}$
 $\therefore H_0$ may be accepted

Confidence Interval (CI)

Confidence Interval is an interval ~~with~~ within which true value of parameter is expected to lie.

① 95% confidence interval ^(or limits) for population

(i) Mean : $\bar{x} \pm 1.96 \frac{\sigma}{\sqrt{n}}$

(ii) Proportion : $p \pm 1.96 \sqrt{\frac{pq}{n}}$

(iii) Difference of means : $(\bar{x}_1 - \bar{x}_2) \pm 1.96 \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$

(iv) Difference of proportions :

$$(p_1 - p_2) \pm 1.96 \sqrt{\frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}}$$

② For 99% confidence intervals, replace 1.96 by 2.58

Ex ① A random sample of 400 items is found to have a mean 82 & S.D. of 18. Find 95% confidence limits (or interval) for the mean of population from which sample is drawn.

Solⁿ $\bar{x} = 82, \sigma = 18, n = 400$

95% confidence limits for population mean

are given by $\bar{x} \pm 1.96 \frac{\sigma}{\sqrt{n}} = 82 \pm 1.96 \frac{18}{\sqrt{400}}$
 $= 82 \pm 1.764$

$\therefore C.I = (80.236, 83.764)$

Ex ② A sample of 200 items from machine A showed that 15 were defective while a sample of 100 items from machine B showed 12 are defective. Find 99% confidence interval for the difference in proportion of defective items from machine A & B.

Solⁿ Given $p_1 = \frac{x_1}{n_1} = \frac{15}{200} = 0.075, q_1 = 1 - p_1 = 0.925$
 $p_2 = \frac{x_2}{n_2} = \frac{12}{100} = 0.12, q_2 = 1 - p_2 = 0.88$

99% confidence interval for difference of proportion

is $(p_1 - p_2) \pm 2.58 \sqrt{\frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}} = -0.045 \pm 0.0966$
 $= (-0.1416, 0.052)$

Test for Small Samples ($n \leq 30$)

① Student's t-test

Let $x_1, x_2, x_3, \dots, x_n$ be a random sample of size n drawn from normal population with mean μ .

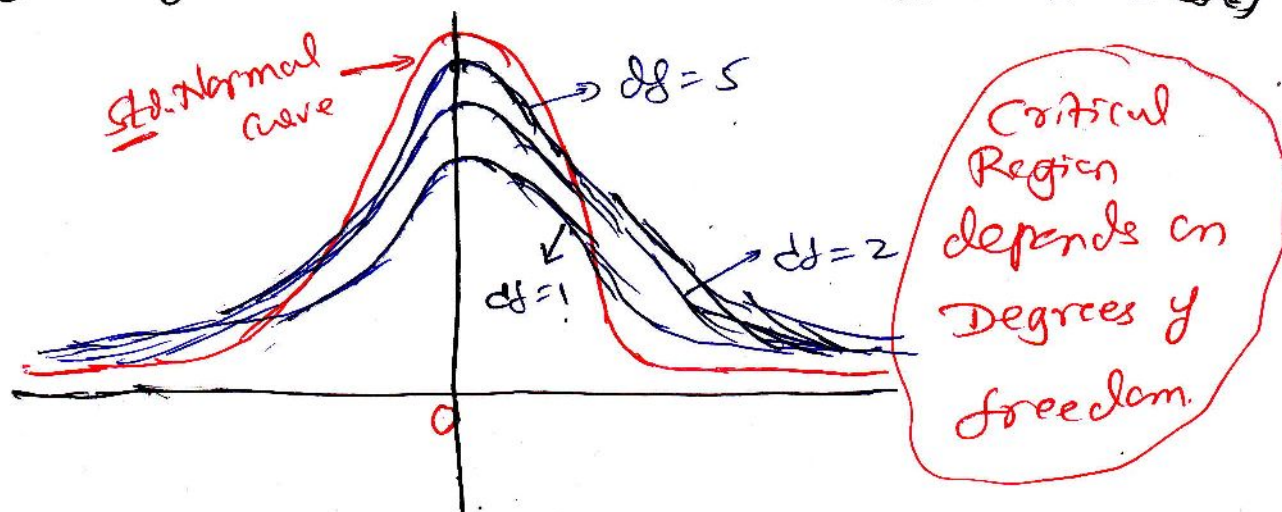
student t-statistic: $t = \frac{\sqrt{n-1} (\bar{x} - \mu)}{s}$

where $\bar{x}$ = sample mean = $\frac{\sum x}{n}$

s = sample S.D. = $\sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2}$

Degrees of freedom: Number of independent variables that a statistic has is called degrees of freedom (df).

For the data with n -observations, degrees of freedom = $df = n - 1$ (for t-test)



Confidence Interval (limits) for the population

mean: $\mu = \bar{x} \pm t_{\alpha} \frac{s}{\sqrt{n-1}}$

Ex ① Ten individuals are chosen at random from a population and their heights in inches are found to be 63, 63, 66, 67, 68, 69, 70, 70, 71, 71. Test the hypothesis that the mean of the universe is 66 inches. Use $t_{0.05} = 2.262$ for 9 df.

Solⁿ Let H_0 : Mean of universe $\mu = 66$
 H_1 : Mean of universe $\mu \neq 66$ (Two tailed)

We have $\sum x = 678$, $\sum x^2 = 46050$, $n = 10$

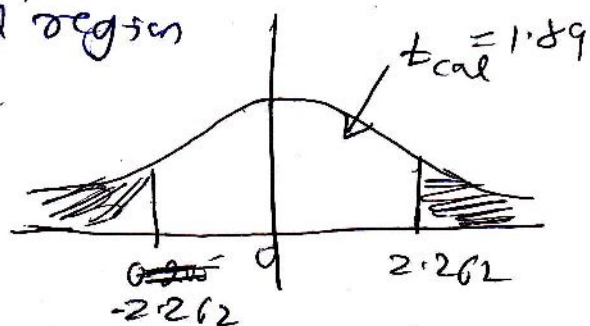
$$\therefore \bar{x} = \frac{\sum x}{n} = 67.8 \quad s = \sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2} = 2.8566$$

Test Statistic: $t = \frac{\sqrt{n-1} (\bar{x} - \mu)}{s} = 1.89$

Since $t_{cal} = 1.89 \notin$ critical region

$\therefore H_0$ may be accepted

$\therefore$ Mean of universe is 66 inches.



(17)

Ex 2 A certain stimulus administered to each of the 12 patients resulted in the following change in blood pressure 5, 2, 8, -1, 3, 0, 6, -2, 1, 5, 0, 4. Can it be concluded that the stimulus will change the blood pressure? Use $t_{0.05}(11) = 2.201$.

Solⁿ Let
 H_0 : The stimulus will not change blood pressure: $\mu = 0$
 H_1 : Stimulus will change blood pressure: $\mu \neq 0$ (Two-tailed)

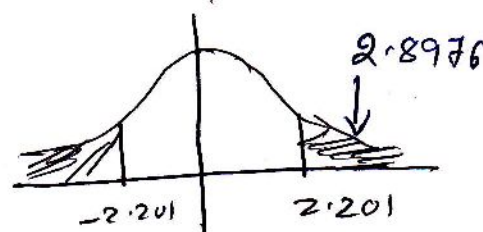
we have

$$\sum x = 31, \quad \sum x^2 = 185, \quad n = 12$$

$$\bar{x} = \frac{\sum x}{n} = 2.5833, \quad s = \sqrt{\frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2} = 2.9569$$

Test statistic:

$$t = \frac{\sqrt{n-1} (\bar{x} - \mu)}{s} = 2.8976$$



$\therefore t_{cal} = 2.8976 \in \text{critical region}$

$\therefore H_0$ is rejected

$\therefore$ stimulus will change the blood pressure.

Ex ③ A machinist makes engine parts with axle diameter of 0.7 inch. A random sample of 10 ~~no~~ parts shows mean diameter of 0.742 inch with S.D. of 0.04 inch. On the basis of this sample would you say ~~that~~ the work is inferior at 5% level of significance, given $t_{0.05}(9) = 2.262$

Solⁿ H_0 : Work is not inferior: $\mu = 0.7$
 H_1 : work is inferior: $\mu \neq 0.7$ (Two-tailed)

we have

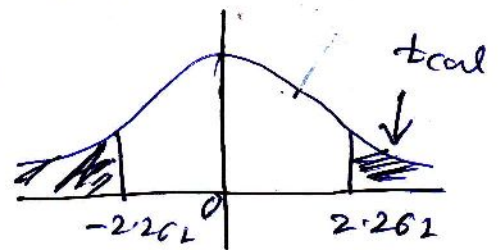
$$\bar{x} = 0.742, s = 0.04, n = 10$$

$$\text{Test statistic } t = \frac{\sqrt{n-1} (\bar{x} - \mu)}{s} = 3.15$$

Since $t_{cal} = 3.15 \in \text{critical region}$

$\therefore H_0$ is rejected

$\therefore$ work is inferior.



Ex ④ A sample of 12 measurements of the diameter of metal ball gave the mean 7.38 mm with S.D. 1.24 mm. Find 95% & 99% confidence limits for actual diameter. Given $t_{0.05}(11) = 2.2$ and $t_{0.01}(11) = 3.11$.

Solⁿ Given $n=12$, $\bar{x}=7.38$, $s=1.24$

① 95% confidence limits for actual diameter

$$= \bar{x} \pm t_{0.05} \frac{s}{\sqrt{n-1}} = 7.38 \pm (2.2) \frac{1.24}{\sqrt{11}} = 7.38 \pm 0.8225$$

$\therefore$ Actual diameter $\in (6.5575, 8.2025)$

② 99% confidence limits for actual diameter

$$= \bar{x} \pm t_{0.01} \frac{s}{\sqrt{n-1}} = 7.38 \pm (3.11) \frac{1.24}{\sqrt{11}} = 7.38 \pm 1.1627$$

$\therefore$ Actual diameter $\in (6.2173, 8.5427)$

Ex 5 Find 95% confidence limits for the population mean given the following frequency distribution of a sample. ($t_{0.05} = 2.13$)

Ex 5	$x:$	2	4	6	8	10
	$f:$	1	4	6	4	1

Solⁿ Given $\sum f = N = 16$, $\sum fx = 96$, $\sum fx^2 = 640$

$$\therefore \bar{x} = \frac{\sum fx}{\sum f} = 6, \quad s = \sqrt{\frac{\sum fx^2}{N} - \left(\frac{\sum fx}{N}\right)^2} = 2$$

$\therefore$ 95% confidence limits for μ is

$$\bar{x} \pm t_{0.05} \frac{s}{\sqrt{N-1}} = 6 \pm (2.13) \frac{2}{\sqrt{15}} = 6 \pm 1.0999$$

$\therefore \mu \in (4.9001, 7.0999)$

t-test for equality of means

Test statistic: $t = \frac{\bar{x} - \bar{y}}{s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$

where $\bar{x} = \frac{\sum x}{n_1}$, $\bar{y} = \frac{\sum y}{n_2}$ $\left| \begin{array}{l} s_1 = \sqrt{\frac{\sum x^2}{n_1} - \left(\frac{\sum x}{n_1}\right)^2} \\ s_2 = \sqrt{\frac{\sum y^2}{n_2} - \left(\frac{\sum y}{n_2}\right)^2} \end{array} \right.$
 $s^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}$

Ex 1 A group of 10 boys fed on a diet A & another group of 8 boys fed on a different diet B for a period of 6 months recorded the following increase in weights

Diet A: 5 6 8 1 12 4 3 9 6 10

Diet B: 2 3 6 8 10 1 2 8

Test whether diets A & B differ significantly regarding their effect on increase in weight.

Use $t_{0.05}(16) = 2.12$

Solⁿ H_0 : Two diets do not differ significantly
: $\mu_1 = \mu_2$

H_1 : Two diets differ significantly : $\mu_1 \neq \mu_2$ (Two Tailed)

we have, $n_1 = 10$, $n_2 = 8$, $\sum x = 64$, $\sum y = 40$

$$\sum x^2 = 512, \sum y^2 = 282$$

$$\therefore \bar{x} = \frac{\sum x}{n_1} = 6.4 \quad \bar{y} = 5$$

$$s_1 = \sqrt{\frac{\sum x^2}{n_1} - \left(\frac{\sum x}{n_1}\right)^2} = 3.2 \quad s_2 = 3.2016$$

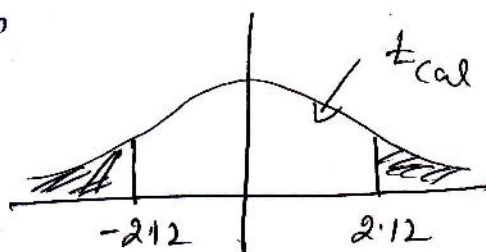
$$\therefore s^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2} = 11.525 \quad \therefore s = 3.395$$

Test statistic: $t = \frac{\bar{x} - \bar{y}}{s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{1.4}{3.395 \times 0.473} = 0.87$

Since $t_{cal} \notin$ critical region

$\therefore H_0$ may be accepted

Hence two diets don't differ significantly.



Ex (2) Two horses A & B were tested according to the time (in sec) to run a particular race with the following results

Horse A: 28, 30, 32, 33, 33, 29, 34

Horse B: 29, 30, 30, 24, 27, 29

Test whether you can discriminate between the two horses. Use $t_{0.05}(11) = 2.2$

Solⁿ H_0 : We cannot discriminate between two horses: $\mu_1 = \mu_2$

H_1 : we can discriminate between two horses: $\mu_1 \neq \mu_2$ (Two-tailed test)

We have $n_1 = 7, n_2 = 6, \Sigma x = 219, \Sigma y = 169$

$$\Sigma x^2 = 6883, \Sigma y^2 = 4787$$

$$\therefore \bar{x} = \frac{\sum x}{n_1} = 31.3, \quad \bar{y} = \frac{\sum y}{n_2} = 28.2$$

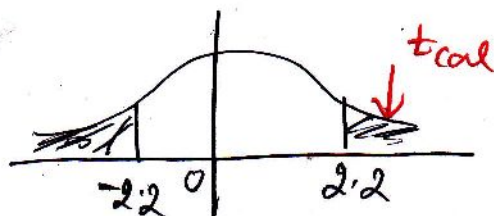
$$s_1 = \sqrt{\frac{\sum x^2}{n_1} - \left(\frac{\sum x}{n_1}\right)^2} = 1.896, \quad s_2 = \sqrt{\frac{\sum y^2}{n_2} - \left(\frac{\sum y}{n_2}\right)^2} = 1.61$$

$$\therefore s^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2} = 3.7 \quad \therefore s = 1.924$$

$$\text{Test statistic: } t = \frac{\bar{x} - \bar{y}}{s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{3.1}{1.924 \times 0.556} = 2.898$$

Since $t_{cal} \in \text{critical region}$

$\therefore H_0$ is rejected



Hence there is a discrimination between 2-horses.

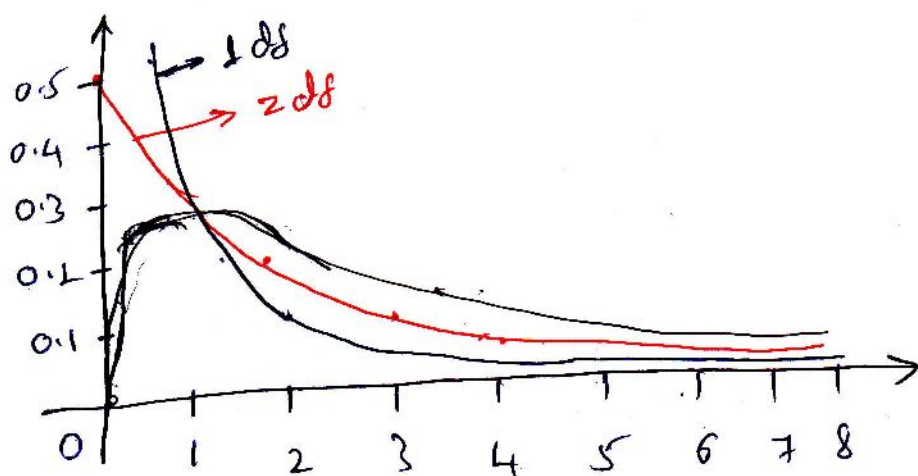
② Chi-Square test (χ^2 -test)

Chi square test is used for testing the significance of difference between the observed values & expected values.

If $O_1, O_2, O_3, \dots, O_n$ are set of observed values & $E_1, E_2, E_3, \dots, E_n$ are set of expected values obtained under some hypothesis, then

Test statistic for testing H_0 is

$$\chi^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i} ; \text{df} = (n-1)$$



Note: Expected values (frequencies) should not be less than 5.

Ex ① A set of $\overset{5}{\nearrow}$ similar coins are tossed 320 times and the observations are

No of heads: 0 1 2 3 4 5

Frequency: 6 27 72 112 71 32

Test the hypothesis that the data follows a binomial distribution (For 5 df, $\chi^2_{0.05} = 11.07$)

Solⁿ H_0 : Data follows binomial distribution

H_1 : Data does not follow binomial distribution

Let X : No of heads, $n=5$

p = prob. of getting head = $\frac{1}{2}$, $q = 1-p = \frac{1}{2}$

$\therefore$ PMF: $P(x) = {}^nC_x p^x q^{n-x} = {}^5C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{5-x} = {}^5C_x \frac{1}{2^5}$

Next

$x = 0, 1, 2, 3, 4, 5$

<u>X</u>	<u>O_i</u>	<u>E_i = 320 x P(x)</u>	<u>(O_i - E_i)² / E_i</u>
0	6	10	1.6
1	27	50	10.58
2	72	100	7.84
3	112	100	1.44
4	71	50	8.82
5	32	10	48.4
			<u>78.68</u>

$\therefore$ Test statistic: $\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i} = 78.68$

Given critical value: $\chi^2_{0.05}(5) = 11.07$

Since $\chi^2_{\text{cal}} > \chi^2_{0.05}(5) \therefore H_0$ is rejected

Hence data does not follow binomial distribution.

Ex 2 Fit a poisson distribution to the following data & test the goodness of fit at 5% level of significance

$x:$	0	1	2	3	4	Given
$f:$	419	352	154	56	19	$\chi^2_{0.05}(3) = 3.82$

Solⁿ H_0 : Poisson distribution is good fit for given data

H_1 : Poisson distribution is not a good fit for given data

We have, Mean = $\frac{\sum fx}{\sum f} = \frac{904}{1000} = 0.904 = \lambda$

$\therefore$ PMF: $p(x) = \frac{e^{-\lambda} \lambda^x}{x!} = \frac{e^{-0.904} (0.904)^x}{x!}, x=0, 1, 2, \dots$

Next

x	O_i	$E_i = 1000 \times p(x)$	$\frac{(O_i - E_i)^2}{E_i}$
0	419	404.95	0.4875
1	352	366.07	0.5408
2	154	165.46	0.7937
3	56	49.86	0.7561
4	19	11.27	5.3019
			7.88

Test statistic: $\chi^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i} = 7.88 > \chi_{0.05}^2(3)$

$\therefore H_0$ is rejected

$\therefore$ Poisson distribution is not a good fit for the given data.

Ex ③ Five dice were thrown 96 times and the number 1, 2 or 3 appearing on the face of the die follows the frequency distribution as follows

No of dice showing 1, 2 or 3 :	5	4	3	2	1	0
Frequency	7	19	35	24	8	3

Test the hypothesis that the data follows binomial distribution. (Use: $\chi_{0.05}^2(3) = 7.815$)

Solⁿ

H_0 : Data follows binomial distribution

H_1 : Data does not follow binomial distribution.

Let x : No of dice showing 1 or 2 or 3

p : Prob. of getting 1 or 2 or 3 = $\frac{3}{6} = \frac{1}{2}$

$q = 1 - p = \frac{1}{2}$, $n = 5$

$\therefore$ PMF: $p(x) = {}^nC_x p^x q^{n-x} = {}^5C_x \frac{1}{2^5}$, $x = 0, 1, \dots, 5$

Next,

<u>X</u>	<u>O_i</u>	<u>E_i = 96 x p(x)</u>	<u>(O_i - E_i)² / E_i</u>
5	7	3	3.556
4	19	15	
3	35	30	0.833
2	24	30	1.2
1	8	15	2.722
0	3	3	
			8.311

Test statistic: $\chi^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i} = 8.311$

Given critical value: $\chi^2_{0.05}(3) = 7.815$

Since $\chi^2_{cal} > \chi^2_{0.05}$ $\therefore H_0$ is rejected

df = 4 - 1 = 3

$\therefore$ Data does not follow binomial distribution.

Ex ④ A die was thrown 264 times and the number appearing on the face (x) follows the frequency distribution

x:	1	2	3	4	5	6
f:	40	30	28	58	54	54

show that the die is biased ($\chi^2_{0.05}(5) = 11.07$)

Solⁿ H_0 : Die is unbiased
 H_1 : Die is biased

Now

<u>X</u>	<u>O_i</u>	<u>$E_i = 264/6$</u>	<u>$(O_i - E_i)^2 / E_i$</u>
1	40	44	0.3636
2	30	44	4.4545
3	28	44	5.8182
4	58	44	4.4545
5	54	44	2.2727
6	54	44	2.2727
			<u>19.636</u>

Test statistic: $\chi^2 = \sum_{i=1}^n (O_i - E_i)^2 / E_i = 19.636$

Given, Critical value: $\chi^2_{0.05}(5) = 11.07$

Since $\chi^2_{cal} > \chi^2_{0.05}(5) \therefore H_0$ is rejected

$\therefore$ Die is biased